On symbolic computations and Post Quantum Cryptography with Lie Geometries.

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Abstract. Assume that the global density of multivariate map over the commutative ring is the total number of its coefficients. In the case of finite commutative ring K with the multiplicative group K^* containing more than 2 elements we suggest multivariate public keys in n variables with the public rule of global density $O(n)$ and degree $O(1)$. Another public keys use public rule of global density $O(n)$ and degree $O(n)$ together with the space of plaintexts $(K^*)^n$ and the space of ciphertext K^n . We consider examples of protocols of Noncommutative Cryptography implemented on the platform of endomorphisms of which allow the conversion of mentioned above multivariate public keys into protocol based cryptosystems of El Gamal type. The cryptosystems and protocols are designed in terms of analogue of geometries of Chevalley groups over commutative rings and their temporal versions.

Keywords. Symbolic Computation, Lie geometries, Noncommutative Cryptography, Multivariate public keys.

1. Introduction

Classical Multivariate Cryptography investigates options of the use of multivariate maps of small degree (usually 2 or 3) defined over finite fields for the construction of public keys. Multivariate Cryptography in wide sense is about cryptographical applications of endomorphisms of multivariate ring $K[x_1, x_2,..., x_n]$ defined over the finite commutative ring K with unity. As usually endomorphism F is given by its values $F(x_i)$, $i=1, 2, ..., n$ written in their standard form. We refer to the maximal number of nonzero coefficients of $F(x_i)$ as the density of F. We propose

(1) public keys with the public rule F from $End\ K[x_1, x_2, ..., x_n]$ of density $O(1)$ and degree $O(1)$ such that the action of F on the space of plaintexts K^n is a bijection. The knowledge of private key allows to decrypt in time $O(n^2)$.

(2) public key with the public rule F from $End\ K[x_1, x_2,..., x_n]$of density $O(1)$ and degree $O(n)$ which induces injective map $(K^*)^n$ to K^n . It has space of plaintexts $(K^*)^n$ and space of potential ciphertexts K^n. The knowledge of private key allows to decrypt in time $O(n^2)$.

(3) implemented secure twisted Diffie-Helman protocol with special generators g and h from $End\ K[x_1, x_2,..., x_n]$ of density $O(1)$ and degree $O(1)$. So Alice and Bob can elaborate common element in time $O(n)$. The security of this algorithm rests on the Conjugacy Power Problem for semigroup $S_n(K)^1$ of elements of $End\ K[x_1, x_2,..., x_n]$ of density $O(1)$.

(4) combinations of Protocol and Public keys as above. Firstly Alice and Bob executes protocol (3) an (4) and elaborate the collision map X from $S_n(K)^1$. Secondly Alice creates public rule F of algorithms (1) and (2). She sends $F+X$ to public user Bob. So he encrypts in time $O(n)$. The knowledge of private key allows her to decrypt.

All elements and subsemigroups of $S_n(K)$ are constructed in terms of special walks on analogue of geometries of Chevalley group $X_m(F)$ where X_m is Coxeter-Dynkin diagram and F is a field defined over the finite commutative ring K with unity.

Some applications of geometries to Communication Theory are already known, If we select the basis in $V=(F_q)^n$ and introduce Hamming metrics on this vector space then each element of projective Geometry which is a subspace of V will be treated as linear code. Since late 80th some other applications of Finite Projective Geometry to Information Security have appeared (see [1], [2], dedicated to Network Coding). In particular, it was used in Cryptography (see [3]) ,where projective geometry were used for authentication protocols or [4], [5] where it was used for the symmetric encryption and key exchange protocols. Finite geometries now are widely used as tools for secret sharing.

Proposed algorithms (1) and (2) are public keys of Multivariate Cryptography (MC) which is one of the fifth directions of Post-Quanum Cryptography designed for the constructions and investigations of asymmetric cryptographic algorithms with the resistance to attacks of the adversary who uses quantum computer. The security of algorithm of MC is based on the complexity of solving the system of nonlinear equations over the finite commutative ring. Traditionally MC uses systems of equations of degree 2 or 3 defined over the finite field (see [6]-[18]). Despite the fact that the last talk on Multivariate Cryptography on Eurocrypt conferences was in 2021 (see [19] and [20]) many new results in this area were obtained (see Proceedings of PQCrypts 2021-2024 and [21]-[23]) and new cryptosystem [23] were submitted to NIST (USA).

Schemes (3)-(4) are based on ideas of Noncommutative Cryptography ([24]-[30]) which is another direction of Postquantum Cryptography. Recent cryptanalytic results on algorithms of Noncommutative Cryptography can be found in [31]-[35].

In Section 2 the constructions of incidence systems which are analogues of geometries of Chevalley group defined over various commutative rings with unity will be presented. Section 3 is dedicated to studies of special ''walks with colour jumps'' on these incidence systems defined over multivariate rings $K[x_1, x_2,\dots, x_n]$ for the design of multivariate transformations of affine spaces K^n and special semigroups of such transformations. In Section 4 new multivariate public keys are introduced. They have better complexity estimates in the comparison with known cryptosystems of Multivariate Cryptography. Twisted Diffie-Hellman protocol of Noncommutative Cryptography which security rests on Conjugace Power Problem is discussed in Section 4. We investigate possible implementations of the protocol with in platform of multivariate transformation of K^n formed by transformation of degree $O(1)$ and density $O(1)$ in Section 5. The implementation with geometry based transformations of stable degree is introduced there. Twisted Diffie Hellman protocol of multivariate nature is used for the design of new multivariate cryptosystems of El. Gamal type. Section 6 is dedicated to illustration of geometrical design of multivariate transformations and semigroups in the case of Coxeter-Dynkin diagram A_{n-1} when Lie Geometry coicedes with the Projective Geometry. Section 7 contains conclusive remarks.

2. On the generalisations of the geometries of Chevalley groups.

The incidence system is the triple (Γ, I, t) where I is a symmetric antireflexive relation (simple graph) on the vertex set Γ, $t : \Gamma\rightarrow N$ is a type function onto the set of types N such that $\alpha I\beta$ and $t(\alpha) = t(\beta)$ implies $\alpha = \beta$. If the cardinality of N is two the terms incidence structure or bipartite graph are used instead of incidence system.

We will use geometries of Coxeter groups and the following bipartite graphs for the construction of larger incidence systems.

Let K be a commutative ring with the unity. Jordan-Gauss graph over K is an incidence structures with partition sets P (points) and L (lines) isomorphic to affine spaces V_1 and V_2 over K such that the incidence relation is given by special quadratic equations over the commutative ring K with unity such that the neighbour of each vertex is defined by the system of linear equation given in its row-echelon form.

Jordan-Gauss graph $J(K)$ of type (s, r) is the special case of linguistic graph of type (s, r) given by the following way (see [45] and further references).

We identify points with tuples of kind $(x)=(x_1, x_2,\dots, x_n,\dots)\epsilon V_1$ and lines with tuples $[y] =[y_1, y_2,\dots, y_n,\dots] \epsilon V_2$. Brackets and parenthesis are convenient to distinguished type of the vertex of the graph. If (x) and $[y]$ are incident $(x)I[y]$ if and only if the following relations hold.

$$a_1x_{s+1}-b_1y_{r+1}=f_1 (x_1, x_2 ,\dots , x_s, y_1, y_2, \dots , y_r),$$
$$a_2x_{s+2}-b_2y_{r+2}=f_2 (x_1, x_2 ,\dots , x_s , x_{s+1}, x_{s+1}, y_1, y_2, \dots , y_r, y_{r+1}), (1)$$
$$\dots$$
$$a_mx_{s+m}-b_my_{r+m}=f_m(x_1, x_2 ,\dots ,x_s, x_{s+1},\dots, x_{s+m-1}, y_1, y_2, \dots , y_r, y_{r+1}, \dots y_{r+m-1})$$

where a_j and b_j, $j=1,2,\dots,m$ are not zero divisors, and f_j are quadratic multivariate polynomials with coefficients from K. We assume that the map sending pair (x), $[u]$ to $(f_1, f_2,\dots, f_m)$ is a bilinear one.

We refer to affine spaces K^s and K^r formed by tuples of kind $(x_1, x_2 ,\dots , x_s)$ and $[y_1, y_2, \dots , y_r]$ as *colour spaces.*.

Let us assume that f_j are given in their standard form , i. e. the list of monomial terms ordered lexicographically.We say that two Jordan graphs $J_1(K)$ and $J_2(K')$ are symbolically equivalent if they are given by the systems of kind (1) with the same number of equations over commutative rings K and K' where quadratic polynomials f_j and f'_j respectively have lists of monomial terms with nonzero coefficients such they differs only by coefficients of monomial terms.

Coxeter geometry is a special case of *group incidence system* $\Gamma(G, G_s)$, $s \in S$. Here G is the abstract group and G_s, $s \in S$ is the family of distinct subgroups of G. The elements of $\Gamma(G, G_s)$, $s \in S$ are the left cosets of G_s in G for all possible $s \in S$. Cosets α and β are incident precisely when $|\alpha \cap \beta| \neq 0$. The type function is defined by $t(\alpha)=s$ where $\alpha = gG_s$ for some $s \in S$.

Let (W, S) be Coxeter system, i.e. W is a group with a set of distinguished generators given by $S=\{s_1, s_2, \dots , s_l\}$ and generic relation $(s_is_j)^{m(i,j)}=e$. Here $M=(m(i,j))$ is a symmetrical l times l matrix with $m(i,i)=1$ and off-diagonal entries satisfying $m(i,j) \geq 2$. Matrix M defines Coxeter-Dynkin diagram D. Let us take $W_i = <S-\{s_i\}>$, $1 \leq i \leq l$ and consider a group incidence system $\Gamma(W)=\Gamma(W, W_i)$, $1\leq i \leq l$. This geometry is called Coxeter geometry of W. Then W_i are referred to as the maximal standard subgroups of W.

Let us consider the definition of BN-pair. Let G be a group**,** B and N subgroups of G, and S be a collection of cosets of $B \cap N$ in N. We call *(G, B, N, S) Tits system* (or we say that G has a *BN-pair*) if

(i) $G=<B, N>$ and $B \cap N$ is normal in N,
(ii) S is a set of involutions which generate $W=N/(B \cap N)$,
(iii) sBw is a subset in $BuB \cap BswB$ for any $s \in S$ and $w \in W$,
(iv) $sBs \neq B$ for all **s** from S.

Properties (1)-(iv) imply that (W, S) is a Coxeter system.

Whenever (G, B, N, S) is Tits system we call the group W by Weyl group of the system or more usually Weyl group of G. The subgroups P_i of G defined by BW_iB are called the *standard maximal parabolic subgroups* of G. The group incidence system $\Gamma(G) = \Gamma(G, P_i)$, $\{1\leq i \leq l\}$ is commonly referred to as *Lie geometry* of G.

Kac – Moody group G is a group with BN- pairs defined via the pair (A, F) where A is generalised Cartan matrix and F is the field. Recall that A is a square matrix with diagonal entries 2 and negative integers in other entries, 0 entries are located symmetrically with respect to main diagonal. It defines Coxeter-Dynkin diagram $X_l=X_l(A)$ via two steps.

(1) Define Kac-Moody Lie Algebra $L(A, F)$.
(2) Take inner automorphism of $L(A, F)$ and construct Kac Moody group $G=G(A, F)$ with the Tits system (G, B, N, S).

Generalised Cartan matrix defines the Weyl group $W(A)$ of the BN-pair G**.**

If $W(A)$ is a finite irreducible Coxeter group then G is Chevalley group over the field F with associated with Coxeter - Dynkin diagram X_l, which belongs to the list A_n, $n\geq 2$, B_n, $n\geq 2$, C_n, $n\geq 2$, D_n, $n\geq 4$, G_2, F_4, E_6, E_7, E_8.

Let us fix Kac-Moody group $G(F)=X_l(F)$ with corresponding Weyl group W. The paper [36Umzh] was dedicated to description of the geometry $\Gamma(G(F))$ in the case of Chevalley group G as algebraic variety in sense of Zariski topology via Jordan-Gauss graphs. Below we introduce the generalisations of geometries of Chevalley groups.

Let (W, S) be the Coxeter system as with $S=\{s_1, s_2, \dots , s_l\}$ and Coxeter- Dynkin diagram D. Assume that $l(g)$ is the length of irreducible representation of g from W in the alphabet S and T is the set of reflections, i. e. elements conjugated with some element from S.

Let $l(\alpha)$ for $\alpha=gW_i$ be the minimal length of the representative of α. With each α we associate $\Delta(\alpha)= \{s \in T: l(s\alpha)<l(\alpha)\}$.

Assume that K is commutative ring with unity, $p(T, K)$ is the set of all partial functions from T to K, i. e. pairs (A, f) where A is a subset of T and f is the function from A to K. Let $L(A)$ be the totality of partial functions with the domain A.

Assume that the empty subset is also on element of $p(T, K)$.

Let $G(W, K^T)$ be the subset of $p(T, K)$ of elements of kind (A, f) where $A=\Delta(\alpha)$. We define type function t on $G(W, K^T)$ via setting $t(A, f))=i$ for A corresponding to $\alpha=gW_i$. Let L_α stands for the set of partial functions $L(A)$ with the domain $A=\Delta(\alpha)$.

For each pair α, β of incident elements of $\Gamma(W)$ we select the Jordan-Gauss graph $J(\alpha, \beta)$ defined over K with the partition sets L_α and L_β and colour spaces $L(\Delta(\alpha))$- $\Delta(\alpha) \cap\Delta(\beta))$ and $L(\Delta(\beta))$- $\Delta(\alpha)\cap\Delta(\beta)$. Let $J_W(K)$ be the variety of selected Jordan-Gauss graphs for each flag (α, β) of the Coxeter geometry.

Finally we define incidence structure $\Gamma(W, J_W(K))$ with the set of elements $G(W, K^T)$ and incidence relation elements $x\epsilon L_\alpha$ and $y\epsilon L_\beta$ are incident if $\alpha I\beta$ in the geometry of Weyl group and xIy in the corresponding Jordan-Gauss graph.

We refer to this incidence system as Jordan-Gauss geometry over Coxeter diagram D.

This is in fact the generalisation of binary case of planar ternary rings Hall's ternary for A_2 defining the projective plane for the general case of arbitrary diagram D of Coxeter system. The following results instantly follows from the interpretation of Lie geometries in the terms of linear Algebra, presented in [36].

Theorem 1.2 *The geometry of Chevalley group over the field F with Weyl group W is isomorphic to some* $\Gamma(W, J_W(F))$.

We say that geometries $\Gamma(W, J_W(K))$ and $\Gamma(W, J'_W(K'))$ are symbolically equivalent if for each flag α, β corresponding Jordan-Gauss graphs are symbolically equivalent.

The examples of $\Gamma(W, J'_W(K'))$ symbolically equivalent to geometries of Chevalley groups can be found in [37] , [38] or [39]. Some of their applications are given in [40] and [39].

The class of $\Gamma(W, J'_W(K'))$ symbolically equivalent to geometry of Theorem 1. 2 is determined by the root system of W.

The incidence systems $\Gamma(W, J_W(K))$ which are symbolically equivalent to geometries of Chevalley groups will be used for the constructions od special elements, groups and semigroups of affine Cremona group $End(K[x_1, x_2, ..., x_n])$ in the next section.

3. Analogues of geometries of Chevalley groups and affine Cremona semigroups.

With each Jordan-Gauss graph $J(K)$ of type s, r given by m equations of kind (1) we associate bipartite Jordan-Gauss graph ${}^nJ(K)$ which is Jordan-Gauss graph over $K[x_1, x_2, ..., x_n]$ with partition sets $K[x_1, x_2, ..., x_n]^{s+m}$, $K[x_1, x_2, ..., x_n]^{r+m}$ and colour space $K[x_1, x_2, ..., x_n]^s$ and $K[x_1, x_2, ..., x_n]^r$ defined by the same system of equations with the coefficients from K. Graphs $J(K)$ and ${}^nJ(K)$ are symbolically equivalent.

Together with family $J_W(K)$ we consider ${}^nJ_W(K)$ obtained via the change of each Jordan graph of the family $J(K)$ for ${}^nJ(K)$. Incidence systems $\Gamma(W, J_W(K))$ and $\Gamma(W, {}^nJ_W(K))$ are symbolically equivalent.

Let us consider some operators on the Jordan-Gauss graph $J(K)$ of type s, r with partition sets V_1 and V_2 of dimensions $s+m$ and $r+m$ respectively.

For each point $x=(x_1, x_2, ..., x_s, x_{s+1}, x_{s+2},..., x_{s+m})$ and symbolic colour $b=(b_1, b_2,..., b_r)$, $b_i\epsilon K[x_1, x_2, ..., x_s]$ from colour space K^r there is a unique neighbour of kind $u=[b_1, b_2,..., b_r, u_1, u_2 ,..., u_m]$. For each line $y=[y_1, y_2, ..., y_r, x_{r+1}, x_{r+2},..., x_{r+m}]$ and colour $b=(b_1, b_2,..., b_s)$ from colour space K^s there is a unique neighbour of kind $u=(b_1, b_2,..., b_s, u_1, u_2 ,..., u_m)$. Let $N_b(v)$ be operator of computing the neighbour of vertex v from $V_1 UV_2$ with the prescribed colour $b \epsilon K^rUK^s$.

We can change the colour of point $x=(x_1, x_2, ..., x_s, x_{s+1}, x_{s+2},..., x_{s+m})$ for the prescribed colour $b=(b_1, b_2,..., b_s)$ from K^s and the colour of$y=[y_1, y_2, ..., y_r, x_{r+1}, x_{r+2},..., x_{r+m}]$ for the colour of $b=(b_1, b_2,..., b_r)$ from K^r. Let $J_b(v)$ be operator of changing the colour of vertex v from $V_1 UV_2$ for the prescribed colour $b \epsilon K^sUK^r$.

We refer to L_α as Schubert cells of the geometry over the diagram and say that L_α is large Schubert cell if α contains Coxeter element of (W, S).

We refer to walk ***w*** of kind $\alpha, \beta, \beta(1),\beta(2), ..., \beta(l), \beta, \alpha$ in the Coxeter geometry as round walk with the edge α, β. We say that the sequence of Schubert cells We refer to the sequence L_α, L_β, $L_{\beta(1)}$, $L_{\beta(2),...}$, $L_{\beta(l)}$, L_β, L_α as Schubert trace of the walk ***w*** and to the sequence of graphs

$J(\alpha, \beta), J(\beta, \beta(1)), ..., J(\beta(l\text{-}1), \beta(l)), J(\beta(l),\alpha)$ as Jordan trace of the round walk w.

Assume that $J(\alpha, \beta)$ has type r, s and defined by m equations. Let us change $\Gamma(W, J_W(K))$ for $\Gamma(W, {}^nJ_W(K))$, $n=m+s$. So $J(\alpha, \beta)$ will be changed for ${}^nJ(\alpha, \beta)$ defined over the ring $R=K[x_1, x_2, ..., x_s, x_{s+1}, ..., x_{s+m}]$.

We form '' the walk with colour jumps '' via selections of colours from the colour spaces of graphs with Jordan trace . The colour of the element is the tuple with coordinates from R but we will use coordinates from the smaller ring $K[x_1, x_2, ..., x_s]$.

Let us assume that $c(\alpha)$ and $n(\beta)$ are possible colours of point and line of ${}^nJ(\alpha, \beta)$, $c(\beta)$ and $n(\beta(1))$ are from colour sets of ${}^nJ(\beta, \beta(1))$, $c(\beta)$ and $n(\beta(1))$ are from colour sets of ${}^nJ(\beta, \beta(1))$, $c(\beta(i)$, $n(\beta(i+1))$, $i=1, 2,..., l\text{-}1$ are pairs of elements of colour spaces of ${}^nJ(\beta(i), \beta(i+1))$, $c(\beta(l)), n(\beta)$ are colours of points and lines of ${}^nJ(\beta(l)), \beta)$, $c'(\beta)$ and $n(\alpha)$ are from colour spaces of $J(\beta, \alpha)$ and $c'(\alpha)$ other colour of the line in $J(\beta, \alpha)$.

The construction.

We take the special point of $J(\alpha, \beta)$ of kind $X=(x_1, x_2, ..., x_s, x_{s+1}, x_{s+2},..., x_{s+m})$ where x_i are generic variables of $K[x_1, x_2, ..., x_s, x_{s+1}, ..., x_{s+m}]$.

We compute $v=J_{c(\alpha)}(X)$, $u=N_{n(\beta)}(v)$ in the graph ${}^nJ(\alpha, \beta)$, $v_1=J_{c(\beta)}(u)$, $u_1=N_{n(\beta(1))}(v_1)$ in the graph ${}^nJ(\beta, \beta(1))$, $v_2=J_{c(\beta(1))}(u_1)$, $u_2=N_{n(\beta(2))}(v_2)$ in the graph ${}^nJ(\beta(1), \beta(2))$, ..., $v_{l\text{-}1}=J_{c(\beta(l\text{-}2)}(u_{l\text{-}2})$, $u_{l\text{-}1}=N_{n(\beta(l\text{-}1))}(v_{l\text{-}1})$ in the graph ${}^nJ(\beta(l\text{-}2), \beta(l\text{-}1))$, $v_l=J_{1c(\beta(l\text{-}1)}(u_{l\text{-}2})$, $u_l= N_{n(\beta(l))})(v_l)$ in the graph ${}^nJ(\beta(l\text{-}1), \beta(l))$, $v_{l+1}=J_{c(\beta(l))}(u_l)$, $u_{l+1}= N_{n(\beta)}(v_{l+1})$ in the graph ${}^nJ(\beta(l), \beta)$, $v_{l+2}=J_{c'(\beta)}(u_{l+1})$, $u_{l+2}=N_{n(\alpha)}(v_{l+2})$ in the graph ${}^nJ(\beta, \alpha)$, $v_{l+3}=J_{c'(\alpha)}(u_{l+2})$ in the graph ${}^nJ(\alpha,\beta)$.

We take the starting point $x=(x_1, x_2,..., x_s, x_{s+1}, x_{s+2},..., x_n)$ and destination point $v_{l+3}=(g_1, g_2, ..., g_s, g_{s+1}, g_{s+2},..., g_n)$ and consider the map F :

$x_1 \rightarrow g_1(x_1, x_2, ..., x_s)$,

$x_2 \rightarrow g_2(x_1, x_2, ..., x_s)$,

...

$x_s \rightarrow g_s(x_1, x_2, ..., x_s)$, (2)

$x_{s+1} \rightarrow g_{s+1}(x_1, x_2, ..., x_n)$,

...

$x_{s+m} \rightarrow g_{s+m}(x_1, x_2, ..., x_n)$.

The map $F= {}^{i,j}F(w, c(\alpha), c(\beta), c(\beta(1)), c(\beta(2), ..., c(\beta(l)),c'(\beta),c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), ..., n(\beta(l)), n'(\beta), n(\alpha))$ depends on the distinct nodes i and j of Coxeter-Dynkin diagram, round walk w**,** and *symbolic trace* $Tr(w)=(c(\alpha), c(\beta), c(\beta), c(\beta(1)), c(\beta(2), ..., c(\beta(l)),c'(\beta),c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), ..., n(\beta(l)), n'(\beta), n(\alpha))$ of w.

We will use also notation ${}^{i,j}F(w,Tr(w))$ for the map F. Let $c'(\alpha)=(g_1, g_2,..., g_s)$ where $g_i \epsilon K[x_1, x_2, ..., x_s]$. We denote the endomorphism $x_i \rightarrow g_i(x_1, x_2,..., x_s)$ as $e(\alpha, \beta, Tr(w))$ and refer to it as rotation of the symbolic trace $Tr(w)$.

Let ${}^{i,j}S(\Gamma(W, J_W(K))$ be subsemigroup of Cremona semigroup $End[x_1, x_2, ..., x_s, x_{s+1},..., x_n]$ generated by all transformations as above with fixed nodes i and j.

We can consider larger subsemigroup ${}^{i,j}S^*(\Gamma(W, J_W(K))$ generated by all elements from semigroups of kind ${}^{i,j}S(\Gamma'(W, J_W(K))$ where $\Gamma'(W, J_W(K))$ is symbolically equivalent to $\Gamma(W, J_W(K))$.

Semigroup ${}^{i,j}S^*(\Gamma(W, J_W(K))$ contains large group of invertible elements ${}^{i,j}G^*(\Gamma(W, J_W(K))$ which contains elements of kind (2) corresponding to round walks of kind $\alpha, \beta, \alpha, \beta, \dots, \beta, \alpha$ with $c'(\alpha)=(g_1, g_2, \dots, g_s)$ such that $x_i \to g_1(x_1, x_2, \dots, x_s), i=1, 2, \dots, n$ is an element of affine Cremona group $Aut\ K[x_1, x_2, \dots, x_s]$.

Elements of symbolic traces are tuples of elements from $K[x_1, x_2, \dots, x_s]$ written in their standard forms. We define degree $deg(u)$ of tuple u as maximum of degrees of its coordinates.

Recall that multivariate map F of K^n onto K^n is given in its standard form of kind $x_i \to f_i(x_1, x_2, \dots, x_n), i=1, 2, \dots, n$ where polynomials f_i are given in the form of the some of monomial terms listed in the lexicographical order. We assume that monomial term M is written in the form $a(x_1)^{t(1)}(x_2)^{t(2)}\dots(x_n)^{t(n)}$, where $t(i)$ are elements of Z_m**,** where m is the order of multiplicative group K^*.

The density of multivariate polynomial is the number of its monomial terms. We define $den(u)$ as the maximal density of its coordinates.

We will use the restriction on degrees and densities of traces to select special elements and subsemigroups of ${}^{i,j}S^*(\Gamma(W, J_W(K))$. For examples we can consider the semigroups of ${}^{i,j}S^*(\Gamma(W, J_W(K))^1$ of all transformations from ${}^{i,j}S^*(\Gamma(W, J_W(K))$ of density and degree of size $O(1)$.

Assume that W is the Weyl group with diagram A_n, B_n, C_n, D_n.

We say that round walk $\alpha, \beta, \beta(1), \beta(2), \dots, \beta(l), \beta, \alpha$ in the geometry of W has finite type if l is finite the difference of types of elements from the walk has size $O(1)$. The following statement follows instantly from definitions.

Proposition 1. 3. [39].*Let* $F(w, Tr(w))$ *be an element of* ${}^{i,j}S^*(\Gamma(W, J_W(K))$ *such that* w is a *round walk of finite type and each coordinate of* $Tr(w)$ *has density and degree of size* $O(1)$ *then the standard form of this transformation is an element* ${}^{i,j}S^*(\Gamma(W, J_W(K))^1$.

Let K be a commutative ring with nontrivial multiplicative group K^*. Let us consider the semigroup $S_n(K)^1$ of ${}^{n}CS(K)= End\ K[x_1, x_2, \dots, x_n]$ consisting of all elements of finite degree and density $O(n)$.

Let us consider enveloping semigroup for ${}^{i,j}S^*(\Gamma(W, J_W(K))$. We can take colours $c(\alpha), c(\beta), c(\beta), c(\beta(1)), c(\beta(2), \dots, c(\beta(l)), c'(\beta), c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), \dots, n(\beta(l)), n(\beta), n(\alpha)$ with coordinates from $K[x_1, x_2, \dots, x_n]$ in the definition of the symbolic trace $Tr(w)$ and define transformation $G(w, T(w))$ of $K[x_1, x_2, \dots, x_n]$. Let ${}^{i,j}ES^*(\Gamma(W, J_W(K))$

be the subsemigroup generated by elements of kind $G(w, T(w))$

and ${}^{i,j}ES^*(\Gamma(W, J_W(K))^1$ stands for the ${}^{i,j}ES^*(\Gamma(W, J_W(K)) \cap S_n(K)^1$ where n is cardinality of $\Delta(\alpha)$.

Remark 1.3. *We can change the subsemigroups* ${}^{i,j}S^*(\Gamma(W, J_W(K))$ *and* ${}^{i,j}S^*(\Gamma(W, J_W(K))^1$ *for the* ${}^{i,j}ES^*(\Gamma(W, J_W(K))^1$ *in the Proposition 1.*

Let $G_n(K)^1=S_n(K)^1 \cap {}^{n}CG(K)$ stands for the subsemigroup of all invertible endomorphisms from ${}^{n}CS(K)$ of finite degree $O(1)$ and density $O(1)$. Notice that the inverse of element of $G_n(K)^1$ can be of large order and large density. So it can be an element outside of $G_n(K)^1$.

Thus we introduce the subgroup $G_n(K)^{1,1}$ consisting of elements with the inverses of finite degree and density $O(1)$.

We can create invertible element $h=F(w, Tr(w))$ from ${}^{i,j}S^*(\Gamma(W, J_W(K))$.For this purpose we can use walk of kind $w= \alpha, \beta, \alpha, \beta, \dots, \beta, \alpha$ with the symbolic trace $c(\alpha), c(\beta), c(\beta), c(\beta(1)), c(\beta(2), \dots, c(\beta(l)), c'(\beta), c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), \dots, n(\beta(l)), n'(\beta), n(\alpha)$ such that $c'(\alpha)=(g_1, g_2, \dots, g_s)$ and endomorphism $e(\alpha, \beta, Tr(w))$: $x_i \to g_i(x_1, x_2, \dots, x_s), i=1, 2, \dots, n$ is an element of $Aut\ K[x_1, x_2, \dots, x_s]$ [39] .We refer to the transformation $F(w, Tr(w))$ as periodical element of ${}^{i,j}S^*(\Gamma(W, J_W(K))$ with *periodical symbolic trace***.**

Let ${}^{i,j}G^*(\Gamma(W, J_W(K))$ stands for the group of elements from ${}^{i,j}S^*(\Gamma(W, J_W(K))$ generated by elements with periodical traces.

We assume that ${}^{i,j}G^*(\Gamma(W, J_W(K))^{1} = {}^{i,j}S^*(\Gamma(W, J_W(K))^{1} \cap {}^{i,j}G^*(\Gamma(W, J_W(K))$.

We refer to the piece of information T as *trapdoor accelerator* of element F from $CG_n(K)$ if the knowledge of T allows us to solve the equation $F(x)=b, b\epsilon K^n$ in polynomial time.

Note that F possess a trapdoor accelerator T then for $LFL_1=G$ where L and L_1 are elements of $AGL_n(K)$ the triple (L, L_1, T) is a trapdoor accelerator. We say about the trapdoor accelerator of complexity $O(n^t)$ if it allows reimage computation intime $O(n^t)$.

Example. Let L be an element of $AGL_n(K)$. Then L^{-1} is the trapdoor accelerator of L of complexity $O(n^2)$.

Proposition 2. 3 [39]. *Let $F= F(w, Tr(w))$ be an element from ${}^{i,j}G^*(\Gamma(W, J_W(K))$ with the rotation $e(\alpha, \beta, Tr(w))$ which has trapdoor accelerator. Then the pair $(Tr(w), T)$ is the trapdoor accelerator of F.*

Proposition 3.3 [39] *Let F be an element of kind $F^{i,j}= F(w, Tr(w))$ from ${}^{i,j}G^*(\Gamma(W, J_W(K))^1$ with the rotation $e=e(\alpha, \beta, Tr(w))$ with trapdoor accelerator of complexity $O(s^2)$. Then F possess trapdoor accelerator T of complexity $O(n)$.*

Proposition 4. 3.[39]. *Let F be an element of kind $F= F(w, Tr(w))$ from ${}^{i,j}G^*(\Gamma(W, J_W(K))^1$ with the rotation $e=e(\alpha, \beta, Tr(w))$* from $e\epsilon G_s(K)^{1,1}$. *Then F is automorphism from $G_n(K)^{1,1}$.*

In the simplest case $e(\alpha, \beta, Tr(w)))$ as in Proposition 3 can be taken of degree 1**.** In fact for each parameter s and degree d we can construct and element from $G_s(K)^{1,1}$ of degree d with its inverse. (see [41] e-Print Archive]).

We refer to elements $G_1, G_2,...., G_k$ from $End\ K[x_1, x_2,..., x_n]$ of degrees $d_1, d_2, ..., d_k$ as stable generators if the maximal degree of elements from the subsemigroup $< G_1, G_2,...., G_k >$ is $max(d_1, d_2,..., d_k)$.

Proposition 5. 3 [39].

Assume that $G_i=F(w_i, Tr(w_i))$, $i=1, 2,..., k$ are elements of ${}^{i,j}S^*(\Gamma(W, J_W(K))$ such that rotations $e(\alpha, \beta, Tr(w_i)$ are *endomorphisms of degree 1. Then they are stable generators of the subsemigroup of ${}^{i,j}S^*(\Gamma(W, J_W(K))$.*

Theorem 1. 3 [39]. Let W be the Weyl group with diagram A_n**,** the semigroup ${}^{i,j}S^*(\Gamma_d(W, J_W(K))$ be subsemigroup generated by elements of kind

${}^{i,j}F(w, c(\alpha), c(\beta), c(\beta(1)), c(\beta(2), ..., c(\beta(l)), c'(\beta), c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), ..., n(\beta(l)), n'(\beta), n(\alpha))$ for which $deg\ c(\alpha)+ deg\ n(\beta)\leq d$,

$deg\ c(\beta)+deg\ n(\beta(1))\leq d)$, $deg\ c(\beta(1)+ deg\ n(\beta(2)) \leq d$, ..., $deg\ c(\beta(l\text{-}1))+deg\ n'(\beta(l))\leq d$, $deg\ c'(\beta)+deg\ n(\alpha) \leq d$, $deg\ c'(\alpha)=1$, where d, $d \geq 2$ is a constant. Then maximal degree of element of ${}^{i,j}S^*(\Gamma_d(W, J_W(K))$ is d.

Theorem 2. 3 [39]. Let W be the Weyl group with diagram A_n, and F be the element of ${}^{i,j}S^*(\Gamma(W, J_W(K))$ of kind ${}^{i,j}F(w, c(\alpha), c(\beta), c(\beta(1)), c(\beta(2)), ..., c(\beta(l)), c'(\beta), c'(\alpha), n(\beta), n(\beta(1)), n(\beta(2)), ..., n(\beta(l)), n'(\beta), n(\alpha))$ for which the maximum of $deg\ c(\alpha)+ deg\ n(\beta)$, $deg\ c(\beta)+deg\ n(\beta)$, $deg\ c(\beta(1)+ deg\ n(\beta(2)$, ..., $deg\ c(\beta(l\text{-}1)+deg n(\beta(l)$, $deg\ c'(\beta)+deg\ n(\alpha)$, $deg\ c'(\alpha)$ is a constant d, $d \geq 2$. Then degree of element F is d.

4. On geometry based cryptosystems of Multivariate Cryptography.

Cryptosystem C1.

Alice selects elements $b_1=F^{i,j(1)}$, $b_2= F^{i,j(2)}$, . .., $b_k=F^{i,j(k)}$ satisfying to Proposition 3 and transformations b_{k+1}, b_{k+2},..., b_{k+m} from $AGL_n(K)$ of density $O(1)$. She selects word $B= b_{i(1)}{}^{k(1)} b_{i(2)}{}^{k(2)} ...b_{i(l)}{}^{k(l)}$ where $1\leq i(j)\leq k+m$, $j=1, 2,..., l$. Alice computes the standard form of B in time $O(n)$ and announces it as a public rule.

Public user Bob encrypts in time $O(n)$. The knowledge of the trapdoors of b_i, $i=1, 2,..., k+m$ allows Alice to decrypt in time $O(n^2)$.The security of this public key rests on the general problem of computation of the reimage of element from $G_n(K)^{1,1}$.

Remark 1. If nonlinear generators b_i have linear rotations then the degree of transformation B is bounded by the product of degrees of $b_{i(j)}$, $j=1, 2,...,l$.

Remark 2. In case of generators from b_i, $i=1, 2,...., k+m$ from $G_n(K)^{1,1}$ Alice can generate pairs (b_i), $(b_i)^{-1}$, $i=1, 2, ..., k+m$ and use the information of them to decrypt in time $O(n)$.

Cryptosystem C2.

Alice generates some pairs (b_i), $(b_i)^{-1}$, $i= 1\ 2, ..., k+m$ from $G_n(K)^{1,1}$. She computes the standard form of $B= b_{i(1)}^{k(1)} b_{i(2)}^{k(2)} ...b_{i(l)}^{k(l)}$and sends it to Bob. He can encrypt in time $O(n)$. Alice is able to decrypt in time $O(n)$.

Let ${}^nES(K)$ be the subsemigroup of $End\ K[x_1, x_2, ..., x_n]$ formed by transformations of kind h: $x_i \to q_i(x_1)^{a(i,1)}(x_2)^{a(i,2)} ...(x_n)^{a(i,n)}$, $i=1, 2,..., n$ where $q_i \epsilon K^*$.

We say that h is multiplicatively invertible if there is $h^* \epsilon\ {}^nES(K)$ such that hh^* and h^*h act on K^* as the identity transformation**.** An algorithm of constructions of pairs (h, h^*) is presented in [42]. We refer to h^* as multiplicative inverse of h. We say that h is of finite type if $deg(h)$ is $O(1)$.

In [42] the following cryptosystem C was suggested.Alice creates several pairs of kind (h_i, h^*_i), $i=1, 2,..., k$ from ${}^nES(K)$. She computes the standard form of $G=G_1G_2,...G_l$, where $G_i \epsilon \{h_1, h_2,..., h_k, h^*_1, h^*_2,..., h^*_k\}$ and sends it to Bob. He writes the plaintext p from $(K^*)^n$. He computes the ciphertext as $G(p)$ in time $O(n^2)$.The knowledge of decomposition of G into the word in alphabet $\boldsymbol{G_i}$ allows Alice to decrypt. She computes $v_l =G^*_l(p)$, $v_{l-1}=G^*_{l-1}(v_l)$,..., $v_1=G^*_1(v_2)=p$ in time $O(n^2)$.

Cryptosystem C3

Alice selects Coxeter-Dynkin diagram X_n and data to work with ${}^{i,\,j}S^*(\Gamma(W, J_W(K))$ and papamerers i, k and $j(1), j(2), ..., j(k)$ to define elements $b_1=F^{i,\, j(1)}$, $b_2= F^{i,\, j(2)}$, ..., $b_k=F^{i,\, j(k)}$ satisfying to Proposition 3 and selects transformations b_{k+1}, b_{k+2},..., b_{k+m} from $AGL_n(K)$.

Alice selects word $B= b_{i(1)}^{k(1)} b_{i(2)}^{k(2)} ...b_{i(l)}^{k(l)}$ where $1\le i(j) \le k+m$, $j=1, 2,..., l$. She computes the standard form of B. Let $m=m(n)$ be the cardinality of $\Delta(\alpha)$, $\alpha\epsilon\Gamma_i(W)$ containing Coxeter element. Alice works with semigroup ${}^mES(K)$. She uses the algorithm of generating pairs (h_i, h^*_i), $i=1, 2,..., k$ from this semigroup. Alice forms $G=G_1G_2,...G_l$, where $G_i \epsilon \{h_1, h_2,..., h_k, h^*_1, h^*_2,..., h^*_k\}$. She computes the standard form of $Y=GB$ of density $O(1)$ and degree $O(n)$. Alice sends the standard form of Y to Bob. He forms the plaintext p from $(K^*)^m$**.** He computes the ciphertext $Y(p)=c$ in time $O(m(n)^2)$ and sends it to Alice. She computes $B^{-1}(c\ )={}^1c$ accordingly to procedure of Cryptosystem 1. The tuple ${}^1c$ is an element of $(K^*)^{m(n)}$. Alices uses the procedure of cryptosystem $\boldsymbol{C}$ of complexity $O(m(n))^2$ to restore p as $G^*({}^1c)$.

Cryptosystem C4.

Alice selects the parameters n and K. She selects the data for the Cryptosystem C and Cryptosystem 2**.** She generates maps G and B. Alices uses the combination of cryptosystem C and Cryptosystem 2. Bob encrypts element p of $(K^*)^n$ with the standard form of GB. Alice restores p via application of the procedures of computation $B^{-1}G^*(c\ )$ for the ciphertext c. Both encryption and decryption procedure take time $O(n^2)$.

5. On twisted Diffie-Hellman protocol with the platform $S_n(K)^1$ and related cryptosystems.

5. 1. The protocol.

Assume that Alice selects element G from $S_n(K)^1$. She generates the pair of elements (H, H^{-1}) from $G_n(K)^{1,1}$ and sends the triple H, H^{-1}, G to Bob via open channel. Alice selects positive integers $r(A)$ and $k(A)$. She computes the standard form of the transformation $H^{r(A)}G^{k(A)}H^{-r(A)} = X(A)$. Alice sends the standard form of $X(A)$ to Bob.

He selects his own private parameters $r(B)$ and $k(B)$, computes $X(B)= H^{r(B)}G^{k(B)}H^{-r(B)}$ and sends it to Alice.

Finally Alice computes the collision transformation X as $H^{r(A)}X(B)^{k(A)}H^{-r(A)}$ and Bob gets $X= H^{r(B)} X(A)^{k(B)}H^{-r(B)}$.

Remark 1. 5. Adversary has to use symbolic computations to express ***H(A)*** (or ***H(B))*** in the form $H^xG^yH^{-x}$ for some parameters x and y (case of Conjugacy Power problem for semigroup generated by G and H from $S_n(K)^1$).

Remark 2. 5. Composition of two elements of density $O(1)$ and finite degree takes time $O(n)$. Execution of the protocol takes $O(1)$ multiplications. And its complexity is $O(n)$. Note that transformation X has density $O(1)$ and finite degree.

Remark 3. 5. The case when period of G and order of H are large numbers is preferable.

Remark 4. 5. Correspondents can take the string of nonzero coefficients of multivariate map X ordered lexicographically and use this string in the alphabet K-$\{0\}$ or its part as a password for encryption algorithm or password for access to resources of information system.

Remarks on cryptanalysis.

Despite the fact that Conjugacy Power Problem in the case of $S_n(K)^1$ is *NP*-hard Alice and Bob have to avoid the general position case when $deg\ (Y^k)= (deg\ Y)^k$. We can see that if G and H and H^{-1} are in general position and have degrees ***a, b*** and ***c.***

Adversary gets $deg\ G^{k(A)}$ of degree $a^{k(A)}$, $deg\ H^{r(A)}$ as $b^{r(A)}$ and $H^{-k(A)}$ as $c^{r(A)}$. Adversary knows degree d of $H^{-r(A)}G^{k(A)}H^{r(A)}$.

So he/she obtains Diophantine equation $a^{k(A)}+b^{r(A)}+c^{r(A)}=d$. So adversary has finite number of options for the pair $k(A)$, $r(A)$. It means that the protocol can be broken in a polynomial time.

Assume that H and G are stable generators of $<H>$ and $<G>$ respectively . Then d is bounded from above by $a+2b$ independently on the choice of $k(A)$ and $r(A)$.

If a and b are stable generators of $<G,H>$ then d is bounded by $max\ (a, b)$ independently on the choice of $k(A)$ and $r(A)$.

So these two cases can be used for the implementation of Twisted Diffie-Hellman protocol.

5.2. On geometry based implementation of the twisted Diffie-Hellman protocol.

We implement both discussed above safe schemes with geometry based transformations of the affine space.

1) Alice selects transformation g of kind $F(w, Tr(w))$ from ${}^{i,j}S^*(\Gamma(W, J_W(K))^1$ and create pair of elements h, h^{-1} from ${}^{i,j'}S^*(\Gamma(W, J_W(K))^1$ as discussed above. She takes two pairs of mutually inverse affine transformation of L, L^{-1} and $L_1, (L_1)^{-1}$ of density $O(1)$ from $AGL_{m(n)}(K)$.

Alice forms transformations $H=LhL^{-1}$, $H^{-1}=Lh^{-1}L^{-1}$ in their standard forms together with transformations $G=L_1g(L_1)^{-1}$. She sends it to her trusted partner Bob via the open channel.

Alice selects positive integers $r(A)$ and $k(A)$. She computes the standard form of the transformation $H^{r(A)}G^{k(A)}H^{-r(A)} = X(A)$. Alice sends the standard form of $X(A)$ to Bob.

Bob selects his own private parameters $r(B)$ and $k(B)$, computes $X(B)= H^{r(B)}G^{k(B)}H^{-r(B)}$ and sends it to Alice.

Finally Alice computes the collision transformation X as $H^{r(A)}X(B)^{k(A)}H^{-r(A)}$ and Bob gets $X= H^{r(B)} X(A)^{k(B)}H^{-r(B)}$.

2) Alice can select $j=j'$ and $L=L_1$. As in this case g , h and h^{-1} are stable generators of the subsemigroup in ${}^{m(n)}SC(K)$.

Remark 5. 5. Adversary has to use symbolic computations to express $H(A)$ (or $H(B)$) in the form $H^xG^yH^{-x}$ for some parameters x and y (case of Conjugacy Power problem for semigroup generated by G and H).

Remark 5.6. Composition of two elements of density $O(1)$ and finite degree takes time $O(m(n))$. Execution of the protocol takes $O(1)$ multiplications. So the transformation X has density $O(1)$ and finite degree.

5. 3. El Gamal type cryptosystem E1 based on the protocol.

It is formed by two consecutive steps.

Procedure 1. Alice and Bob executes presented above twisted Diffie-Hellman Protocol end elaborate the collision transformation X.

Alice selects ${}^{i,k}S^*(\Gamma(W, J_W(K))$ and elements L_2 and L_3 from $AGL_n(K)$ of densities $O(1)$ and $O(n)$ respectively.

Procedure 2. Alice can use Proposition 2. She takes invertible element H' of kind $F(w, Tr(w))$ such that the tuples of the trace are degree $O(1)$ from ${}^{i,k}G^*(\Gamma(W, J_W(K))$ with the rotation $e(\alpha, \beta, Tr(w))$ from ${}^{s'}CG(K)$ of and trapdoor accelerator T. The map H' has polynomial density $O(m(n)^\gamma)$ for some γ. She forms $Y= L_2 H' L_3$, computes the standard form of $X+Y$ of density $O(m(n)^{\gamma+1})$ to Bob. This steganographic action is secure.

Bob subtracts X and gets the standard form of Y. Note that he can encrypt in time $O(m(n)^{2+\gamma})$. The knowledge of symbolic trace of w and T allows Alice to decrypt in polynomial time. Security of this cryptosystems rests on the security of the protocol.

Remark 5.7. Let us consider convenient robust version when Alice use pairs h, h^{-1} from $G_{s'}(K)^{1,1}$ and takes F from ${}^{i,k}S^*(\Gamma(W, J_W(K))^1$ **.**She sets $h=e(\alpha, \beta, T)$. In this case the knowledge of L_2, L_3, $Tr(w)$, h^{-1} allows her to decrypt in time $O(m(n)^1)$. Bob encrypts in time $O(m(n)^2)$.

Remark 5.8. During its history researchers of Multivariate Cryptography produce many examples of elements $F\epsilon F_q[x_1, x_2,..., x_t]$ of degree 2 and 3 with the trapdoor accelerators**.** Most of them are broken. The Procedure 1 gives the second life to them. Let $t=m(n)$. Alice and Bob can elaborate element X of prescribed degree from prescribed number of variables. So Alice selects the historical cryptosystem and sends the standard form of $G= L_1FL_2$ to Bob for some invertible L_i of degree 2 (see [43]) .Bob can encrypt and knowledge of T allows Alice to decrypt.

Remark 5.9. Classical multivariate cryptosystems can be used for the design of the rotation map for $F(w, Tr(w))$ with the trapdoor accelerator of Procedure 2. For example one can take quadratic Imai- Matsumoto multivariate map an its trapdoor accelerator in the case of appropriate dimension and the field of characteristic 2 (see [44]).

5.4. On El Gamal type cryptosystem E2.

Below we introduce the alternative option for the Procedure 2.

Procedure 3. Recall that ${}^nES(K)$ is the subsemigroup of $End\ K[x_1, x_2, ..., x_n]$ formed by transformations of kind h**:** $x_i\to q_i(x_1)^{a(i,1)}(x_2)^{a(i,2)} ...(x_n)^{a(i,n)}$, $i=1, 2,..., n$ where $q_i\epsilon K^*$.

Let W be Weyl group and $w=\alpha, \beta, \beta(1)=\alpha, \beta(2)=\beta, ...\beta(l)=\alpha, \beta, \alpha$ is a periodic walk of finite type with odd $l\geq 1$. Assume that Jordan-Gauss graph $J(\alpha, \beta)$ has type s, r and given by $\boldsymbol{t}$ equations, $\boldsymbol{m=s+t}$. We consider the map E of kind

$x_1\to q_1(x_1)^{a(1,1)}(x_2)^{a(1,2)} ...(x_s)^{a(1,s)}$,

$x_2\to q_2(x_1)^{a(2,1)}(x_2)^{a(2,2)} ...(x_s)^{a(2,s)}$,

...

$x_s\to q_s(x_1)^{a(s,1)}(x_2)^{a(s,2)} ...(x_s)^{a(s,s)}$,

$x_{s+1}\to f_1(x_1, x_2 ,..., x_s)+b_{11}x_{s+1}+b_{12} x_{s+2}+...+b_{1t} x_{s+t}$, (3)

$x_{s+2}\to f_2(x_1, x_2 ,..., x_s)+b_{21}x_{s+1}+b_{22} x_{s+2}+...+b_{2t} x_{s+t}$,

...

$x_{s+t}\to f_t(x_1, x_2 ,..., x_s)+b_{t1}x_{s+1}+b_{t2} x_{s+2}+...+b_{tt} x_{s+t}$,

of density $O(1)$ and finite degree such that $E(x_1), E(x_2), ..., E(x_s)$ defines multiplicatively invertible element of ${}^sES(K)$ of finite degree and matrix $(b_{i,j})$ defines invertible linear transformation of K^t. We select the $Tr(w)$ of kind $c(\alpha), c(\beta), c(\beta(1)), c(\beta(2), ..., c(\beta(l)),c'(\beta),c'(\alpha), n(\beta), n(\beta(1), n(\beta(2)), ..., n(\beta(l)), n'(\beta), n(\alpha))$ formed by tuples with coordinates from $K[x_1, x_2,..., x_s]$ such that $c'(\alpha)=(G(x_1), G(x_2), ..., G(x_s))$ where $G =HL$ is the composition of multiplicatively invertible transformation H of degree $O(1)$ from ${}^sES(K)$ and element L of $GL_s(K)$ of density $O(1)$.

So we form $F=F(w, Tr(w))$ and compute the standard form of $G=EFL_1$ where $L_1\epsilon AGL_m(K)$. Note that L is arbitrary element of the affine group. So G has density $O(n)$. It is easy to see that we can compute the reimage of G for the vector b from $G((K^*)^s\cdot K^t)$.

The algorithm is the following. Let usconsider the equation $G(x_1, x_2,..., x_s, x_{s+1}, x_{s+2},..., x_{s+t})=b$ where $x_i\epsilon K^*$ for $i=1, 2,..., s$. We compute $(L_1)^{-1}(b)=(d_1, d_2,..., d_s, d_{s+1},..., d_m)$ and consider the equation $G(x_1, x_2,..., x_s)=(d_1, d_2,..., d_s)$ which is equivalent to $H(x_1, x_2, ..., x_s)= L^{-1}(d_1, d_2,..., d_s)$. The knowledge of H^* allows us to get the solution$x_i=e_i$, $i=1, 2,..., s$.

So we compute colours $d(\alpha)=c(\alpha)(e_1, e_2,..., e_s)$, $d(\beta)=c(\beta)(e_1, e_2,..., e_s)$, $d(\beta(1))=c(\beta(1))(e_1, e_2,..., e_s)$, $d(\beta(2))=c(\beta(2)(e_1, e_2,..., e_s)$, ..., $d(\beta(l))=c(\beta(l))(e_1, e_2,..., e_s)$,$d'(\beta)=c'(\beta)(e_1, e_2,..., e_s)$, $d'(\alpha)=c'(\alpha)(e_1, e_2,..., e_s)$, $m(\beta)=n(\beta) (e_1, e_2,..., e_s)$, $m(\beta(1))=n(\beta(1)(e_1, e_2,..., e_s)$, $m(\beta(2))=n(\beta(2))(e_1, e_2,..., e_s)$, ..., $m(\beta(l))=n(\beta(l))(e_1, e_2,..., e_s)$, $m'(\beta)= n'(\beta)(e_1, e_2,..., e_s)$, $m(\alpha)=n(\alpha) (e_1, e_2,..., e_s))$.

Firstly we change the colour of $(d_1, d_2,..., d_s, d_{s+1},..., d_m)$ and get the vertex $(e_1, e_2,..., e_s, d_{s+1},..., d_m)=u_0$. Secondly we compute the neighbour of this vertex as $v_0=N_{d'(\beta)}(u_0)$. We change of the colour of this vertex and get $u_1=J_{m'(\beta)}(v_0)$. Next we compute the neighbour of $v_1=N_{d(\beta(l))}(u_1)$.

We change the colour of $\boldsymbol{v_1}$ and get the vertex $u_1=J_{m(\beta(l))}(v_1)$. We continue this process and getting the neighbour v of intermediate vertex of colour $d(\alpha)$. We change the colour of v for $(e_1, e_2,..., e_s)$ and get $u=(L_1)^{-1}F(w, tr(w))^{-1}$**.**

Finally we take equations $E(x_1, x_2,....x_m)=u$ and get the solution $(p_1, p_2,..., p_s, p_{s+1}, ..., p_{s+t})$ which is the reimage of b.

We modify presented above cryptosystem E1 based on twisted Diffie-Hellman protocol and introduce cryptosystem E2.

Let K be finite commutative ring with rather large multiplicative group.

We can select finite field or arithmetical ring Z_m, $m>3$ of residues modulo m.

Assume that Alice and Bob selected considered above protocol with the platform $L^{\ i,\ j}ES^*(\Gamma(W, J_W(K))^1L^{-1}$ or $L^{\ i,\ j}S^*(\Gamma(W, J_W(K))^1 L^{-1}$ where W is the Weyl group over the diagram from the list $\{A_n, B_n, C_n, D_n\}$. Assume that difference of types of α and β has size $O(1)$ and $\Delta(\alpha)$ has cardinality m of size $O(n^2)$. Alice and Bob get the collision element X which is a multivariate transformation of the affine space K^m of density $O(1)$ and finite degree d, $d\geq 2$. Alice changes the node j for other node k of diagram such that the difference of i and k is $O(1)$. The case of $k=j$ is also possible. Let γ be the coset of $\Gamma(w)$ of type k which contains Coxeter element.

Alice works with the round walk of kind $\alpha, \gamma, \alpha,\gamma, ... \alpha, \gamma, \alpha$ of length $O(1)$.

She uses Procedure 3 in the case of the graph $J(\alpha, \gamma)$ and generates the transformation G. Independently on the choice of γ the knowledge of w, $Tr(w)$, E and L_1 allows Alice to find the reimage of element from $G((K^*)^{s+t}$. So Alice sends $X+G$ to Bob. He encrypts the tuples from $(K^*)^m$ via the applications of the standard form of G.

6. Illustrative examples.

6. 1. Selected case.

Let us consider the $n\text{-}1$ dimensional projective geometry defined over the field F, i. e. the geometry of Chevalley group $PGL_n(F)$. This is a totality of subspaces of F^n**.** Coxeter system in this cases is a symmetric group with generators $(1, 2), (2,3), ..., (n\text{-}1, n)$. It has a diagram A_{n-1}**.** The geometry of Weyl group can be identified with the totality of nonempty subspaces of $\{1, 2, ..., n\text{-}1, n\}$. Subsets A and B are incident if and only if $A\cap B=A$ or $A\cap B=B$. The cardinality of A is the value of the type function t on this set. Let us consider

Elements $J_k=\{n, n\text{-}1, n\text{-}2,..., n\text{-}k+1\}$ of type k corresponds to coset containing the Coxeter element.

Then $\Delta(J_k)$ contains transpositions

$(1, n\text{-}k+1), (2, n\text{-}k+1,..., (n\text{-}k, n\text{-}k+1),$

$(1, n\text{-}k+2) , (2, n\text{-}k+2), ..., (n\text{-}k, n\text{-}k+2),$

...

$(1, n)\quad , (2, n),\quad ...\quad , (n\text{-}k, n)$

and $\Delta(J_{k-1})$ consists of

$(1, n\text{-}k+2, (2, n\text{-}k+2),...., (n\text{-}k, n\text{-}k+2), (n\text{-}k+1 , n\text{-}k+2),$

$(1, n\text{-}k+3), (2, n\text{-}k+3), ...(n\text{-}k, n\text{-}k+3), (n\text{-}k+1, n\text{-}k+3),$

...

$(1, n),\quad (2, n), ...,\qquad (n\text{-}k+1, n).$

It is easy to see that colour spaces are F^{n-k} and F^{k-1}.

We set $n\text{-}k=m$ and $k\text{-}1 =l$ and identify $\Delta(J_k)$ with $\{1, 2,..., m, 11, 12, ..., 1m, 21, 22,..., 2m,...., ml, l2,..., ml\}$ and $\Delta(J_{k-1})=\{1, 2,..., l, 1l, 12,..., m1\}$.

Incidence between $x=(x_1, x_2, ..., x_m, x_{11}, x_{12},..., x_{ml})$ and $y=[y_1, y_2,..., y_l, y_{11}, y_{12},...,y_{ml}]$ the corresponding Jordan Gauss graph will be given by equations

$x_{ij}\text{-}y_{i,j}=x_iy_j,\ i=1, 2, ..., m, j=1, 2,..., l.$

We assume that m and l are parameters of size $O(n)$. So the number of variables is $O(n^2)$. Note that $m+l=n\text{-}1$ which is the dimension of projective geometry.

6. 2.The implementation of Cryptosystem 1 of El Gamal type.

Assume that Alice and Bob decided to use the described above cryptosystem and work in single class of symbolically equivalent Jordan-Gauss graphs containing presented above graph.

So they select finite commutative ring with unity K and plan to work with the temporal graph with partition set K^{m+ml}, K^{l+lm} and incidence xIy given by equations

$\alpha(i,j)x_{ij}\text{-}\beta(i, j)y_{i,j}=\gamma(i, j)x_iy_j,\ i=1, 2, ..., m, j=1, 2,..., l.$

where $\alpha(i,j)$ and $\beta(i, j)$ are elements of the commutative group K^* of the ring K and $\gamma(i, j)\neq 0.$

So Alice takes the round walk w of kind $A, B, A, B,..., A, B, A$ where $A= J_k$ and $B= J_{k-1}$.

She selects colours $c(\alpha)$, $c(\beta)$, $c(\beta)$, $c(\beta(1))$, $c(\beta(2), ..., c(\beta(l)), c'(\beta), c'(\alpha)$, $n(\beta)$, $n(\beta(1)$, $n(\beta(2))$, ..., $n(\beta(l))$, $n'(\beta)$, $n(\alpha)$ of density $O(1)$ and of finite degrees with coordinates from $K[x_1, x_2,..., x_{m+ml}]$ in the definition of the symbolic trace $Tr(w)$ and define transformation $G(w, T(w))$ of $K[x_1, x_2,..., x_{m+ml}]$. During each step of the computation of neighbour in the graph Alice can change parameters $\alpha(i,j)$, $\beta(i, j)$ and $\gamma(i, j)$.

She computes $F=F(w, Tr(w))$ of degree $O(1)$ and of density $O(1)$. Alice selects transformation L such that densities of L and L^{-1} are of density $O(1)$ and computes the standard form of $G=LFL^{-1}$.

After that she selects round walk w' of kind $A, B, A, B,..., A, B, A$ of odd length l'. Alice forms symbolic trace of w' as $Tr(w')=(c'(\alpha), c'(\beta), c'(\beta), c'(\beta(1)), c'(\beta(2), ..., c'(\beta(l')), c''(\beta), c''(\alpha), n'(\beta), n'(\beta(1), n'(\beta(2)), ..., n'(\beta(l')), n''(\beta), n'(\alpha))$ formed by the tuples with coordinates from $K[x_1, x_2,..., x_m]$ such that the tuple $c''(\alpha)=(g_1, g_2, ..., g_m)$ defines automorphism $x_i \rightarrow g_i(x_1, x_2,..., x_m)$, $i=1, 2,...., m$ of $K[x_1, x_2,..., x_m]$ with the density $O(1)$.

She computes $h=F(w', tr(w'))$ and the standard form of $H=LhL^{-1}$.Alice delivers H and G. So correspondents executes described above twisted Diffie-Hellman protocol with these generators.

Assume that they elaborate the collision endomorphism X of $End(K[x_1, x_2,..., x_m])$ of degree d.

Alice forms nonlinear bijective transformations $Y=L_1h'L_2$

where $h'=F(w'', Tr(w''))$ is the transformation with w'' of kind $A, B, A, ..., B, A$ and *sparce* trace $Tr(w'')$ formed by tuples with coordinates of density $O(1)$, L_1 is an element of density $O(1)$ from $AGL_n(K)$, $L_2 \epsilon AGL_n(K)$, $n=m+lm$. Alice can control the degree of h and h' via selection of symbolic traces satisfying to conditions of Theorem 1 or Theorem 2.The density of Y will be of size $O(n)$. So Bob can encrypt in time $O(n^2)$.

Alice sends $X+Y$ to Bob. So Bob can use the standard form for encryption and Alice can decrypt because of her knowledge of L_1, L_2 and $Tr(w'')$.

6.3. The implementation of Cryptosystem 2 of El Gamal type.

Alice create transformation E of K^{m+ml} of kind 3 of density $O(1)$ with $s=m$. She can take the same round walk w' modify its trace via the change of $c''(\alpha)$ for the $(B(x_1), B(x_2), ..., B(x_m))$ where B is the composition of element E' from ${}^mEG(K)$ of degree $O(1)$ and L' from $AGL_m(K)$ of density $O(1)$. Let ${}^1Tr(w')$ stands for the modified trace.

Alice forms Y' as the standard form of $EF(w', {}^1Tr(w))L_2$. Alice sends $X+Y'$ to her partner.

So Bob can use Y' for the encryption of tuples from $(K^*)^{m+ml}$. The knowledge of E, w', ${}^1Tr(w')$ and L_2 allows Alice to decrypt in time $O(n^2)$, $n=m+ml$.

Example.

Notice that the history of multivariate public keys is a source of maps with the trapdoor accelerator. In [43] the following renovation of bijective Imai-Matsumoto cryptosystem were suggested. Which is the pair (F, T) where F is bijective quadratic transformation of $(F_q)^n$, $q=2^s$ and T is a tiple L_1, L_2 , t where L_i are elements from $AGL_n(q)$, t is the parameter of kind $q^r +1$ which is mutually prime with q^n-1. Correspondents executes Twisted Diffie-Hellman protocol as in Corollary 1 in the case affine space $(F_q)^n$ and elaborate the collision map X. Alice keeps the trapdoor accelerator T and sends the standard form $X+F$.

6. 4. On the conversion of multivariate public key to Cryptosystem of El Gamal type.

Assume that endomorphism F of $K[x_1, x_2,..., x_n]$ is used as the public rule of some Multivariate Cryptosystem MC.Alice and her trusted partner Bob can select the platform subsemigroup of ${}^nCS(K)$ and execute some protocol of Multivariate Cryptography and elaborate the collision map $\boldsymbol{X}$. So Alice sends $X+F$. Bob encrypts with F and Alices uses her private key to decrypt. We can use this scheme with the Twisted Diffie-Hellman protocol for the encryption. So we convert cryptosystems *C1, C2, C3* and *C4* to the protocol based systems *EC1, EC2, EC3* and *EC4*.

7. Conclusion.

Most bijective public keys of Multivariate Cryptography are given by quadratic maps of kind $x_1 \to f_1(x_1, x_2,..., x_n)$, $x_2 \to f_2(x_1, x_2,..., x_n)$, ..., $x_n \to f_n(x_1, x_2,..., x_n)$ defined over finite fields. So public user can encrypt in time $O(n^3)$. The knowledge of private key allows to decrypt in polynomial time.In this paper we use sparce multivariate maps of degree $O(1)$ with $O(n)$ nonzero coefficients for the construction of public keys defined over finite commutative ring K with unity. The degree can be rather large (of size 10^2 or 10^3 particularly) and K may have zero divisors. This situation is unusual for studies in Cryptanalysis. Anyway in case of such algorithms theoretically public user can encrypt in time $O(n)$. We hope that this fact attracts attention of many specialists in Cryptanalytics to proposed algorithms. Moving further in this direction we suggest multivariate cryptosystem working with space of plaintexts $(K^*)^n$ where K^* is multiplicative group of K based on multivariate map F of degree $O(n)$.The restriction of F on $(K^*)^n$ is injective and knowledge of private key makes decryption feasible. In proposed version total number of nonzero coefficients is $O(n)$. So public user can encrypt in time $O(n^2)$ which is better than the case of multivariate quadratic map. Cryptanalytic studies of the case of unbounded by constant degree are unknown.

Additionally we construct protocol based multivariate cryptosystems with option to encrypt in time $O(n)$ in terms of Noncommutative Cryptography with the platform of multivariate transformations. We hope that it stimulates further investigations in the intersection area of Noncommutative and Multivariate Cryptographies. For the design of these cryptosystems we use very special endomorphisms of $K[x_1, x_2,..., x_n]$ defined in terms of analogue of Lie geometries over K.

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